

Steel Plate Faults Exploratory Data Analysis

DN

October 2026

1. Introduction

1.1 Objective

Explore how recorded steel-plate fault types vary with steel grade, plate thickness, defect size and defect geometry using SQL, pandas and Python visualisation.

1.2 Important limitation

The dataset contains fault instances but not the total number of non-faulty plates inspected. The analysis therefore describes patterns among recorded faults; it cannot estimate true defect rates or establish causation.

2. Dataset overview

2.1 Dataset snapshot

Dataset source: Buscema, M., Terzi, S., & Tastle, W. (2010). Steel Plates Faults, UCI Machine Learning Repository (DOI: 10.24432/C5J88N). Accessed via Kaggle: Faulty Steel Plates.

Item	Value
Records	1,941
Fault classes	7
Steel types	A300 and A400
A300 fault records	777
A400 fault records	1,164

2.2 Questions explored

- How do fault types differ between A300 and A400 steel?
- Do different plate thicknesses show different fault distributions?
- Do different fault types have characteristic sizes or shapes?

3. Exploratory analysis

3.1 Fault distribution by steel type

I first compared the percentage distribution of the seven recorded fault categories within each steel type. Percentages are more useful than raw counts here because A300 and A400 contain different numbers of fault records.

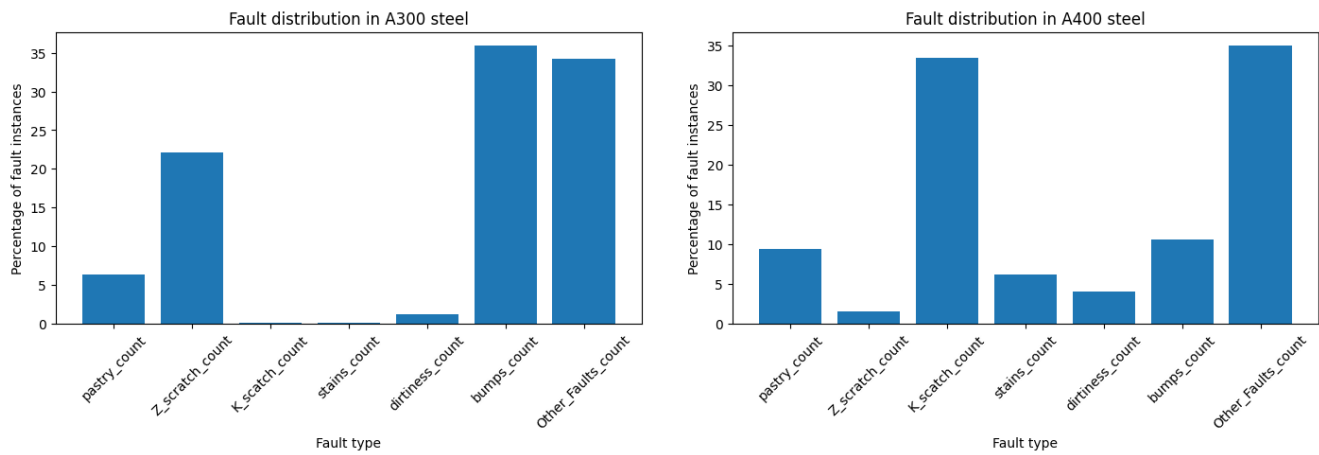


Figure 1. Percentage distribution of recorded fault types within A300 and A400 steel.

Observation: A300 is dominated by Bumps (35.9%) and Other_Faults (34.2%), while A400 has large shares of K_Scratch (33.5%) and Other_Faults (35.0%). These differences describe the available fault records only; they do not show that one steel grade is more fault-prone.

3.2 Plate thickness and fault composition

Fault counts were grouped by plate thickness. Because some thicknesses contain far more records than others, I converted the fault counts within each thickness to percentages and used a heatmap to compare composition.

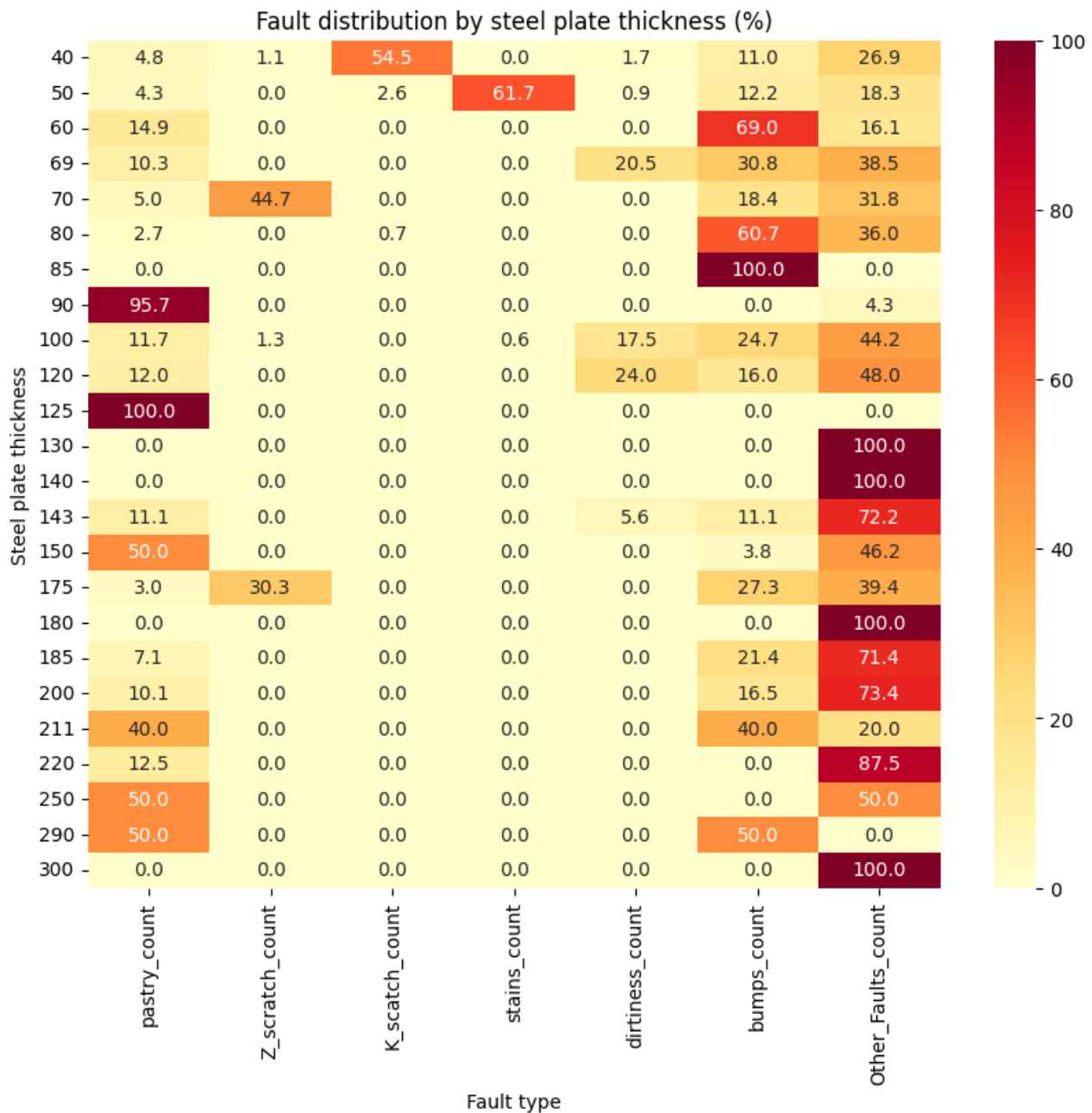


Figure 2. Fault-category percentages by steel plate thickness.

Observation: The mix of recorded fault categories changes across thickness groups. However, several thicknesses have very small sample sizes, so strong conclusions should not be drawn from isolated high percentages.

3.3 Fault size

I used Pixels_Areas as a measure of how much image area each detected defect occupies. A box plot was used because it shows the median, spread and extreme observations for every fault class rather than reducing each class to a single average.

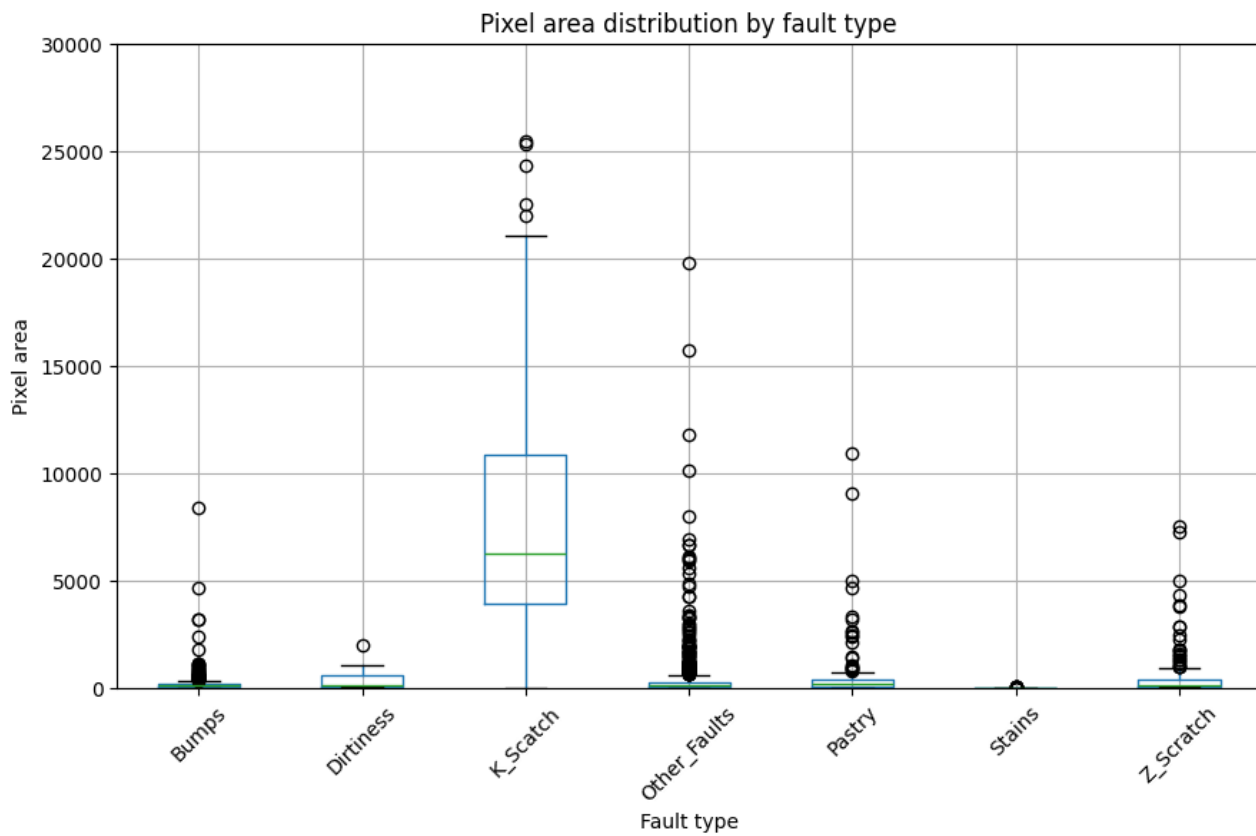


Figure 3. Zoomed pixel-area distribution by fault type.

Observation: K_Scratch stands out clearly: its median pixel area is about 6,281 pixels, compared with roughly 120–209 pixels for most other categories. Stains are particularly small, with a median of 16.5 pixels. Several classes contain large outliers, which were retained rather than removed without evidence that they are errors.

3.4 Fault geometry

To move beyond overall area, I derived fault width and height from the X and Y coordinate limits. This allowed me to compare the geometric characteristics of the fault categories.

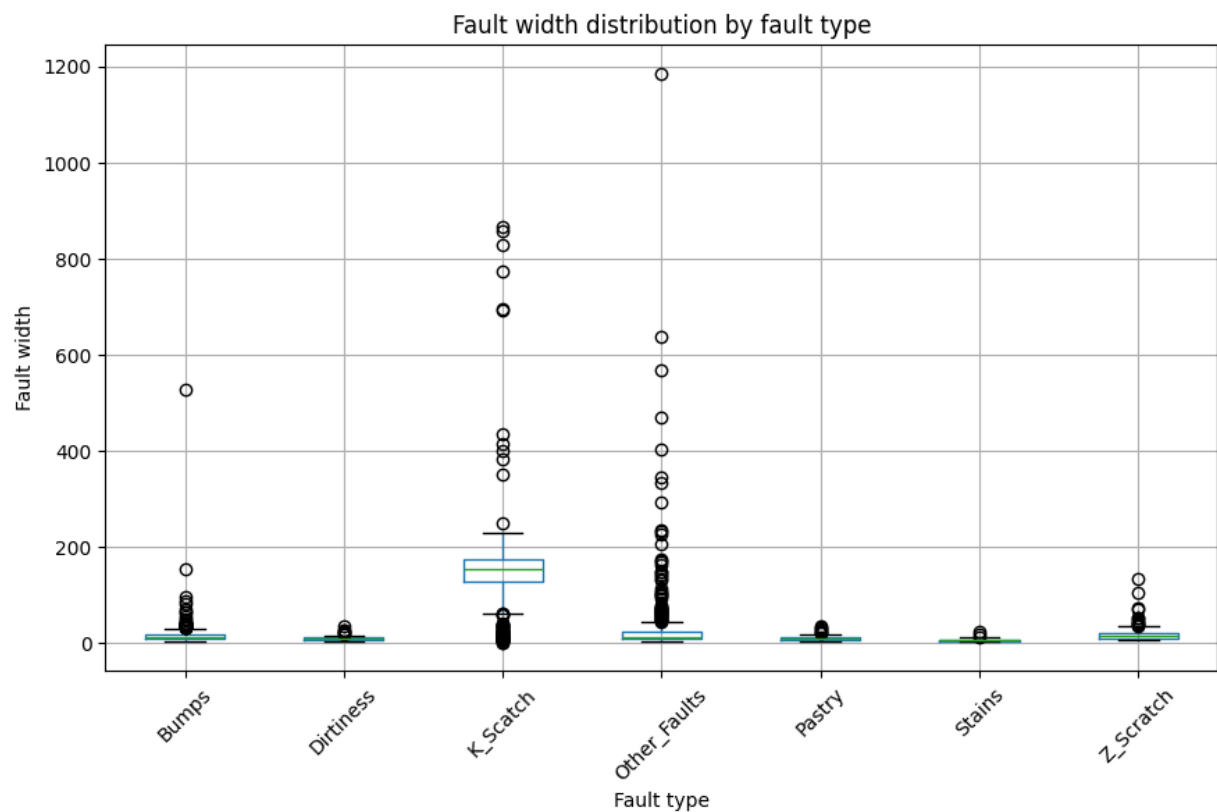


Figure 4. Fault-width distribution by fault type.

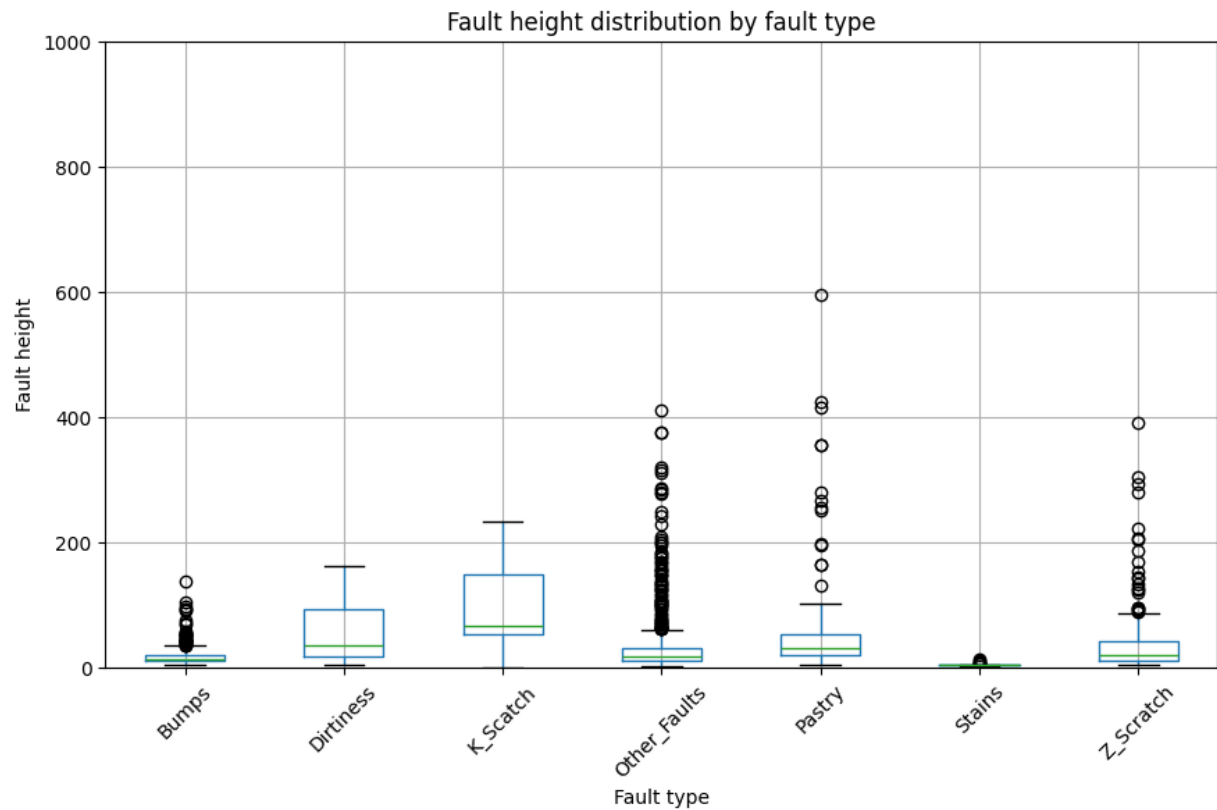


Figure 5. Zoomed fault-height distribution by fault type.

Observation: K_Scratch also has a much larger median width than the other classes. One K_Scratch record has an extreme height of 18,141, while the next highest height is 597. I retained the observation and used a zoomed plot so the typical distributions remained readable.

4. Key findings

- A300 and A400 show different distributions of recorded fault categories.
- Fault observations are unevenly distributed across plate thicknesses, which makes sample size important when interpreting percentages.
- K_Scratch defects are substantially larger in pixel area than most other fault categories.
- Fault categories show different width and height distributions, suggesting that geometry contains useful information for distinguishing fault types.
- The dataset is more suitable for fault-characterisation and classification questions than for estimating manufacturing defect rates or determining causes.

5. Limitations

- The total number of inspected plates is not provided, so true defect rates cannot be calculated.
- Some thickness groups contain very few observations.
- Several variables are image-derived measurements rather than direct manufacturing-process parameters.
- Observed associations should not be interpreted as causal relationships.

6. Reflection

This project reinforced an important point that also applies in engineering analysis: the quality of the conclusions depends on the quality and scope of the available data. I initially wanted to understand whether certain steel types or plate thicknesses were more strongly associated with particular faults. As the analysis progressed, it became clear that the dataset only contained recorded fault instances and did not include the full production or inspection population. This meant I could identify patterns within the available faults, but not calculate true fault rates or make causal statements about material or thickness.

From a data-analysis perspective, the project was useful for working across SQL and pandas in a practical way. I used SQL for grouping and aggregation, and pandas for reshaping data, percentage-based comparisons and distribution analysis. The visualisation stage was particularly useful in deciding how best to present different types of information: bar charts for categorical comparisons, heatmaps for grouped patterns and box plots for variation and outliers.

The dataset also appears well suited to a machine-learning classification problem, where the measured defect characteristics could be used to predict one of the seven fault classes. For this project, however, I chose to stop at the exploratory analysis stage. The main value was in understanding the structure of the data, identifying meaningful patterns, recognising where the evidence was insufficient, and selecting appropriate analytical methods to support the questions being asked.